

Chapter 2

Thinking Like an Economist

This chapter introduces the two hats an economist wears, and the two simple models we will lean on all semester: the circular-flow diagram and the production possibilities frontier.

It closes with a distinction, positive versus normative, that will help you read every policy debate you encounter more clearly.



The Economist as a Scientist

Economists play two different roles: as scientists, they try to explain the world as it is; as policy advisers, they try to improve it.

When they act as scientists, economists use the same scientific method as a physicist or a biologist. They develop a theory, collect data, and see whether the data confirm or refute it.

Economists cannot usually run controlled laboratory experiments on an entire economy. Instead, they rely on natural experiments: real-world events that history provides for free.

Definition. A **natural experiment** is a real-world event that, by chance, exposes one group to a change while leaving a similar comparison group unaffected, creating something close to a controlled experiment.

Example: A War and the Price of Oil

When a war in the Middle East disrupts the supply of crude oil, oil prices spike around the world.

For consumers, that spike is bad news. For policymakers, it poses a hard choice. But for economists, it is a natural experiment: a chance to study how a shock to one key resource ripples through the world's economies.

This is not just a hypothetical. The US-Israel war against Iran, which began in February 2026 and threatened to close the Strait of Hormuz, pushed Brent crude up more than 40 percent within weeks.

Taiwan, which imports nearly all its energy, felt this directly as fuel and electricity costs rose through 2026.

Example: COVID-19 as a Natural Experiment

The COVID-19 pandemic gave economists an especially large natural experiment. Different countries, and even different states within a country, imposed lockdowns of very different length and strictness, often for reasons unrelated to their economies.

Comparing regions that locked down hard with similar regions that did not lets researchers estimate how much of the resulting drop in economic activity came from the policy itself, rather than from the virus alone.

Taiwan is itself a useful point of contrast: it delayed widespread lockdowns until 2021, well after most Western economies, giving economists another natural comparison to work with.

Assumptions and Models

If you ask a physicist how long a marble takes to fall from a building, she will assume away air resistance, since the effect is too small to matter.

Economists simplify in exactly the same spirit. To study international trade, we might assume the world contains only two countries and two goods, so we can see the essence of the problem without its full complexity.

Example. An economic model of Taiwan's electronics exports might assume that every firm behaves like a single representative chipmaker.

The assumption is false in detail, since thousands of smaller suppliers exist, but it lets us see the main forces at work without tracking every firm individually.

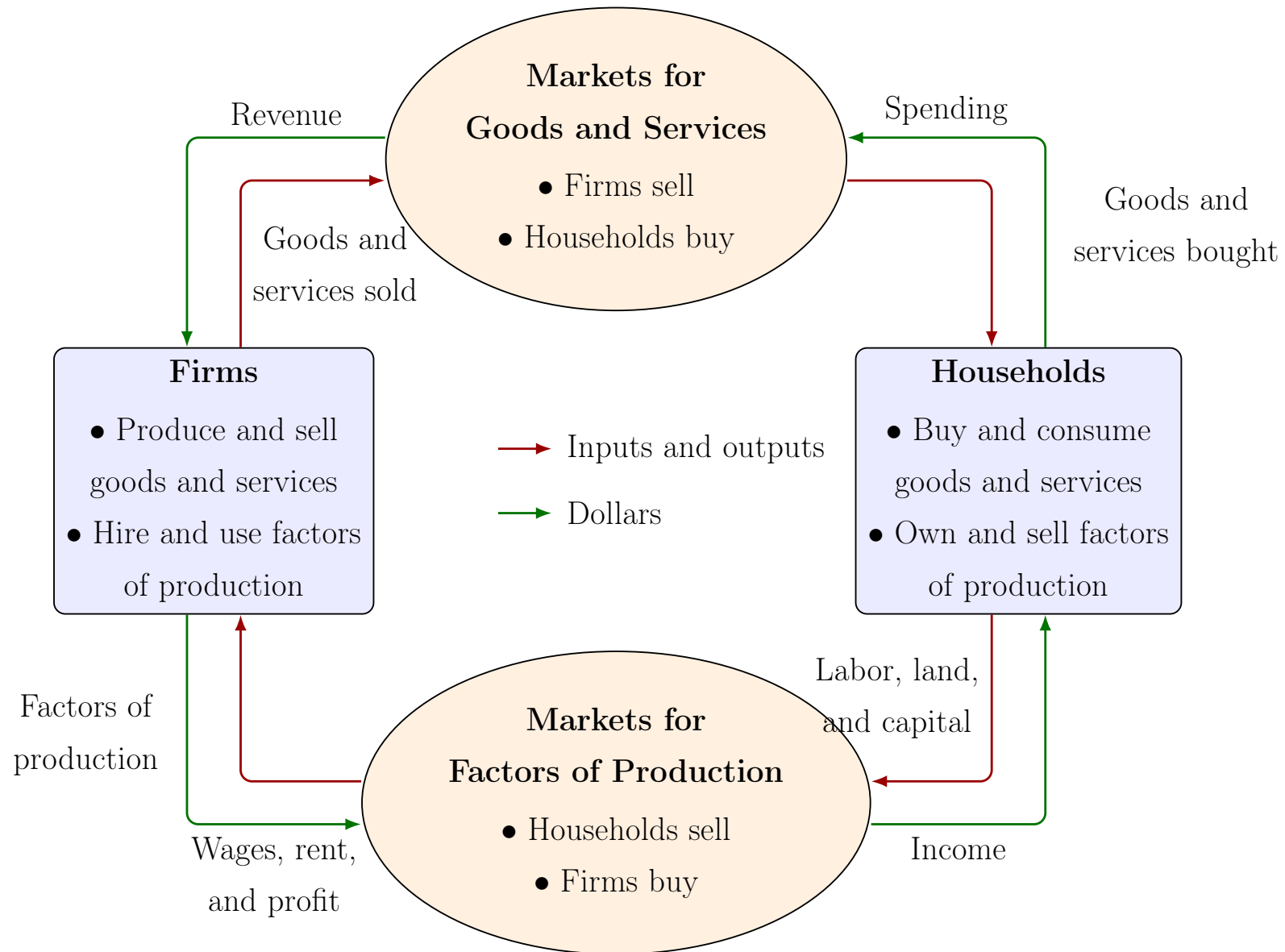
Our First Model: The Circular-Flow Diagram

Definition. The **circular-flow diagram** is a visual model of the economy that shows how dollars flow through markets among households and firms.

The diagram simplifies the economy down to two decision makers, firms and households, interacting in two markets.

One market is for goods and services. The other is for the factors of production, the labor, land, and capital that firms use to produce those goods and services.

The Circular Flow, Drawn Out



Following a Dollar Around the Flow

Imagine a NT\$100 bill starting in your wallet. You spend it at a bubble tea shop, and it becomes revenue for the firm.

The shop then uses that revenue to pay rent to its landlord or wages to its staff. The same NT\$100 becomes income for someone else's household.

That person's household is, once again, holding money it can spend in the market for goods and services. The circular flow starts over.

This simple diagram leaves out government and international trade. But it captures the basic loop that a more realistic model would only complicate.

Our Second Model: The Production Possibilities Frontier

Definition. The **production possibilities frontier** (PPF) is a graph that shows the various combinations of output that an economy can possibly produce, given its available factors of production and available technology.

Unlike the circular-flow diagram, the PPF is built with numbers, which lets it illustrate trade-offs and opportunity cost precisely rather than just in general terms.

Example: Taiwan's Production Possibilities

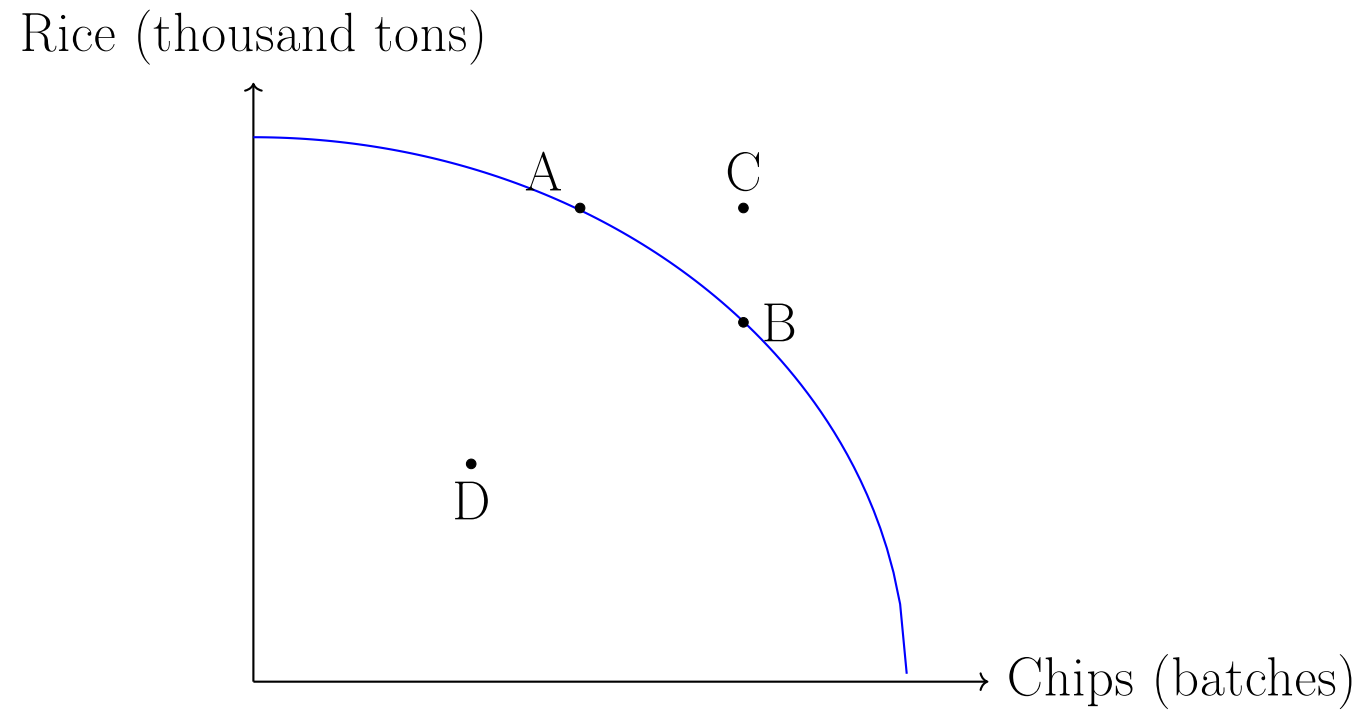
Suppose Taiwan's labor and capital could be devoted entirely to producing either semiconductor chips or rice.

Using all its resources on rice alone, Taiwan could produce 10,000 tons. Using all its resources on chips alone, it could produce 12 batches.

In between, it could produce combinations such as 6 batches of chips and 8,700 tons of rice, or 9 batches of chips and 6,600 tons of rice.

Every one of these combinations, and every combination in between them, lies on Taiwan's production possibilities frontier.

Plotting the Frontier



Discussion: Reading the Frontier

Look at the four points on the previous page. Points A and B sit exactly on the curve; point C sits above and to the right of it; point D sits below and to the left of it.

- Is point C, with more of both chips and rice than A or B, something Taiwan could actually produce right now?
- Is point D, with less of both goods than A, something Taiwan could actually produce?
- If D is possible, why would Taiwan ever want to avoid producing there?

Answer: Efficient, Inefficient, and Infeasible

Definition. An outcome is **efficient** if the economy is getting all it can from its scarce resources: no more of one good can be produced without producing less of the other.

Each point falls into one of three categories:

- Points A and B are **efficient**: they sit on the frontier.
- Point C is **infeasible**: it lies outside the frontier, beyond what Taiwan's current resources and technology allow.
- Point D is **inefficient**: it lies inside the frontier, so some resources, perhaps idle workers or unused factories, are sitting unused.

Taiwan could produce more of both goods simply by putting those resources to work.

Moving Along the Frontier: Opportunity Cost

Moving from point A to point B means shifting resources out of rice farming and into chip production.

Taiwan gains 3 more batches of chips but gives up roughly 2,100 tons of rice to get them. The opportunity cost of those extra chips is measured directly by how far the frontier drops.

Remark. This is exactly Principle 2 from Chapter 1, that the cost of something is what you give up to get it, now expressed as the slope of a graph rather than as a sentence.

Why the Frontier Bows Outward

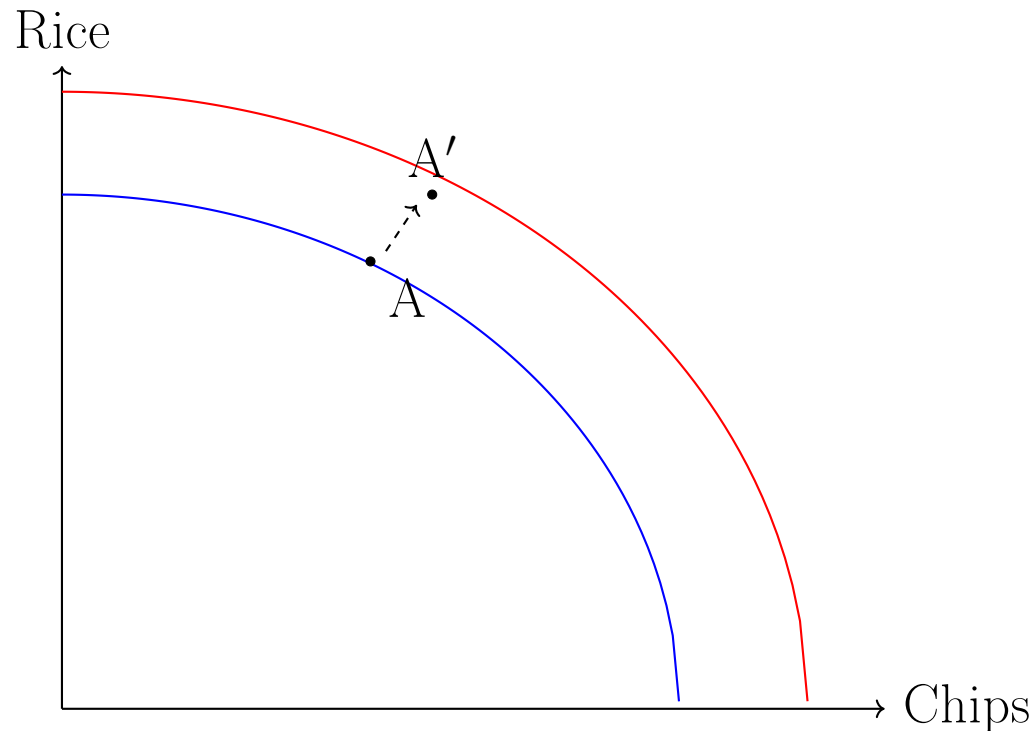
Taiwan's frontier curves rather than running in a straight line, because workers are not interchangeable between rice farming and chip fabrication.

When Taiwan is producing mostly rice, the workers who shift into a handful of chip batches are the ones least suited to farming. The rice given up is small.

Once Taiwan is already producing many batches of chips, squeezing out a few more requires pulling in workers who are relatively good at farming and only mediocre at semiconductor work.

Each additional batch of chips then costs much more rice than the last one did. That rising opportunity cost is what bows the frontier outward.

Economic Growth Shifts the Frontier



A better irrigation technique or a more advanced chip fabrication process lets Taiwan produce more of one or both goods with the same workers, pushing the whole frontier outward. Growth lets Taiwan move from a point like A to a point like A' that was simply not feasible before.

Microeconomics and Macroeconomics

Definition. **Microeconomics** is the study of how households and firms make decisions and how they interact in specific markets.

Definition. **Macroeconomics** is the study of economy-wide phenomena, including inflation, unemployment, and economic growth.

A microeconomist might study why rents in Taipei have risen so quickly. A macroeconomist might study why Taiwan's overall inflation rate has stayed low.

The two fields are closely related, since economy-wide outcomes are built up from millions of individual decisions. But each has its own set of models, and is often taught as its own course.

The Economist as Policy Adviser

When economists explain why a policy would raise or lower unemployment, they are acting as scientists.

When they recommend a policy because it would improve people's lives, they are acting as policy advisers. That recommendation rests on more than facts alone.

Remark. President Harry Truman once said he wanted to find a “one-armed economist,” since every economist he asked for advice answered, “On the one hand ... on the other hand ...”

That habit is not economists being evasive. It reflects Principle 1 from Chapter 1: people face trade-offs. A policy that helps one group, or one generation, usually costs another.

Positive versus Normative Statements

Definition. **Positive statements** are descriptive: they attempt to describe the world as it is, and can in principle be confirmed or refuted with evidence.

Definition. **Normative statements** are prescriptive: they attempt to say how the world ought to be, and cannot be settled by evidence alone.

“A cap on rents reduces the quantity and quality of rental housing available in Taipei” is positive: it is a claim about cause and effect that data could confirm or refute.

“The city government should cap rents in Taipei” is normative: it is a value judgment about what policy should try to achieve.

Discussion: Positive or Normative?

For each statement, decide whether it is positive or normative, and explain how you can tell.

1. Raising the minimum wage increases unemployment among young workers.
2. The government should raise the minimum wage.
3. A higher carbon tax would reduce Taiwan's greenhouse gas emissions.
4. Taiwan should invest more in renewable energy.



Answer: Positive or Normative?

The first and third statements are positive: each describes a cause-and-effect relationship that data could, in principle, confirm or refute, whether or not the claim turns out to be true.

The second and fourth statements are normative: each expresses a judgment about what should be done, which no amount of data alone can settle.

Remark. A statement does not need to be true to be positive. It only needs to be the kind of claim that evidence could confirm or refute.



Economists as Policy Advisers in Taiwan

Economists work throughout Taiwan's government:

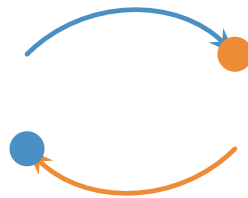
- The Central Bank of the Republic of China (Taiwan) sets monetary policy.
- The National Development Council helps plan long-run economic strategy.
- The Directorate-General of Budget, Accounting and Statistics compiles the data that every other policy discussion relies on.

Being right is only part of a policy adviser's job. Elected officials weigh economists' advice against many other constraints: political, legal, and practical. Even good advice is only one ingredient in the final decision.

Why Economists Disagree

Economists sometimes give conflicting advice for two basic reasons:

1. They may disagree about the validity of competing positive theories, much as scientists in any field can have different hunches about how the world works before the evidence is all in.
2. They may hold different values, and therefore different normative views about what a policy should try to accomplish, even when they agree completely about its likely effects.



What Most Economists Actually Agree On

Despite their reputation for disagreement, economists agree on more than most people expect. Surveys of professional economists find overwhelming majorities behind propositions such as these.

- A ceiling on rents reduces the quantity and quality of housing available.
- Tariffs and import quotas usually reduce a country's overall economic welfare.
- A minimum wage increases unemployment among young and unskilled workers.
- Restricting employers from outsourcing work to other countries makes the economy worse off overall.

Chapter in a Nutshell: Economists as Scientists

Economists try to study the world with a scientist's objectivity, making simplifying assumptions and building models to understand what would otherwise be an overwhelmingly complex reality.

Two of the simplest such models are:

- The **circular-flow diagram**, which shows how households and firms interact through markets.
- The **production possibilities frontier**, which shows the trade-offs a society faces between two goods.

Microeconomics studies individual households and firms; macroeconomics studies the economy as a whole.

Chapter in a Nutshell: Economists as Policy Advisers

A positive statement asserts how the world is. A normative statement asserts how the world ought to be. Only the first kind can be confirmed or refuted with evidence alone.

When economists disagree, it is usually because they hold different scientific judgments, different values, or both.

Economists are united far more often than their public reputation suggests, even though policymakers do not always follow their advice.

Key Concepts

- The circular-flow diagram and the production possibilities frontier are the two basic models introduced in this chapter.
- Efficient, inefficient, and infeasible describe where an outcome sits relative to the production possibilities frontier.
- Microeconomics and macroeconomics describe the two traditional subfields of the discipline.
- Positive statements and normative statements describe the two roles economists play, as scientists and as policy advisers.